

Augmented Hide-and-Seek: Evaluating Spatial Decision-Making and Player Experience in a Multiplayer Location-based Game

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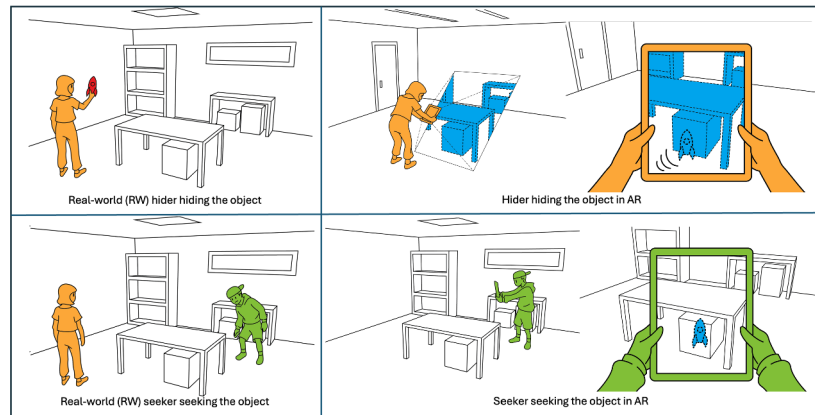


Figure 1: (Left) Real-world hide-and-seek gameplay. (Right) Hide-and-seek remote gameplay in augmented reality.

ABSTRACT

This paper presents a study on the impact of enabling remote gameplay via Augmented Reality (AR) on spatial decision-making and player experience in a hide-and-seek game. We designed a remote multiplayer handheld-based AR game and evaluated how it influences players' spatial decision-making strategies, engagement, and gameplay experience. In a study with 60 participants, we compared remote gameplay in our AR game to traditional hide-and-seek. Our findings show that AR enhances gameplay by improving spatial interactions, decision-making, and collaboration. Despite navigation challenges, AR has the potential to foster player engagement and social interaction, contributing to the design of future AR games and beyond.

Index Terms: Location-based games, spatial decision making, augmented reality, remote gameplay.

1 INTRODUCTION

Location-based games (LBGs) integrate players' physical locations and actions into the gameplay, transforming everyday spaces into interactive environments that blend the physical and virtual worlds for playful exploration and social interaction [6]. AR enhances these experiences by overlaying virtual elements onto real-world (RW) settings, transforming familiar spaces into immersive and interactive playgrounds [1]. However, existing location-based augmented reality games (LBARGs) often underutilize the unique characteristics of RW locations, integrating them only as superficial

or passive elements rather than as core components of gameplay mechanics [3, 5]. Incorporating RW characteristics into LBARGs enables players to explore locations through engaging spatial interactions, enhancing the overall remote multiplayer gameplay experience. Understanding how these characteristics influence player engagement, spatial decision-making, and gameplay experience is crucial for the design of remote multiplayer LBARGs. Therefore, this study explores remote multiplayer AR game design via a hide-and-seek format, investigating how AR affects player interactions with virtual and physical elements in a shared environment. It aims to offer recommendations for AR game developers to enhance engagement, particularly for players connecting from different locations.

2 METHODOLOGY

2.1 Game Design

The game was designed to connect players across remote locations through synchronized, interactive gameplay. It features two primary roles: Hider and Seeker. The Hider can see a 1:1 AR representation of the Seeker's environment, created using the Scaniverse app installed on a LiDAR-enabled iPad, as shown in Figure 2. This virtual representation allows the Hider to place virtual objects, adjusting their position using a simple interface with thumbsticks.

Once the hiding phase is complete, the Seeker uses AR to explore their RW space to locate the hidden virtual objects. Objects appear in the Seeker's physical location as shown in Figure 3. The game allows players to exchange "Hot & Cold" verbal cues, enabling real-time voice communication between players, which mirrors traditional hide-and-seek while accommodating remote play. The Hider can observe the Seeker's position and orientation through a 3D frustum. We also added a countdown timer to introduce time pressure to the seeker, ensuring balanced difficulty while maintaining player engagement.

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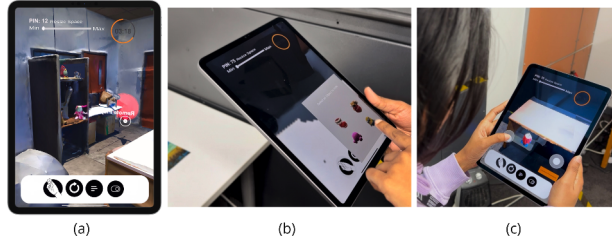


Figure 2: (a) Hider's view of Seeker's space with a 1:1 AR representation. (b) Hider selects an object from the virtual inventory. (c) Hider uses virtual thumbsticks to position the object in the remote space.



Figure 3: Remotely hidden AR objects in the seeker's location.

2.2 User Study Design

A controlled experiment involved 60 participants (ages 18–40; $M = 27.88$, $SD = 6.05$). After providing informed consent, they played two rounds as Hider and Seeker, switching roles afterwards. Participants completed validated questionnaires, including the Game Engagement Questionnaire (GEQ) [2], User Engagement Scale (UES-SF) [7], and User Experience Questionnaire (UEQ-S)[8], and concluded with interviews. Gameplay behaviours and eye-tracking data were recorded. Ethics approval was obtained from the University of Canterbury's Human Research Ethics Committee (HREC 2024/21/LR-PS).

3 RESULTS AND DISCUSSION

AR gameplay allowed participants to make unconventional spatial decisions. Hiders picked vertical spaces (e.g., ceilings, mid-air) to hide objects, which isn't feasible in RW condition, resulting in longer hiding times ($M=68.9s$, $SD=15.4$) compared to RW ($M=48.6s$, $SD=12.2$; $p<0.001$). This indicates that AR significantly influences players' spatial decision-making, fostering more imaginative and unpredictable behavior [4]. This supports prior research emphasizing AR's potential to transform traditional gameplay with new spatial possibilities. Seekers adapted search patterns exhibiting concentrated upward fixations in AR (Figure 4). Despite these complexities, object-finding times remained comparable between conditions, suggesting effective adaptation by Seekers.

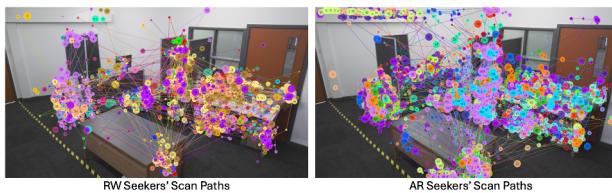


Figure 4: Seekers' scan paths between RW and AR conditions.

While RW gameplay felt more realistic, AR induced greater immersion, with players reporting losing track of time and space (Table 1). Reward scores were significantly higher in AR based on UES-SF. This engagement may stem from the novelty and interactivity of AR, allowing users to manipulate virtual objects in real-world settings. However, players faced challenges with virtual con-

trols when placing virtual objects, contributing to decreased usability. UEQ-S results show that the RW condition scored higher on Pragmatic Quality (2.0 ± 0.5) versus the AR condition (1.6 ± 0.5). However, the AR condition excelled in Hedonic Quality (2.0 ± 0.6) compared to RW (0.6 ± 0.5). Overall, AR (1.8 ± 0.5) outperformed RW (1.3 ± 0.5). This indicates that, despite usability issues, AR's engaging gameplay led to higher scores.

Table 1: Statistically significant results from GEQ and UES-SF

Measure	RW Gameplay (Mean $\pm$ SD)	AR Gameplay (Mean $\pm$ SD)	p-value
GEQ: The game feels real	4.2 $\pm$ 0.6	3.5 $\pm$ 0.7	$p = 0.0001$
GEQ: I lose track of where I am	2.9 $\pm$ 0.9	4.1 $\pm$ 0.8	$p = 0.0189$
GEQ: I feel spaced out	2.5 $\pm$ 0.7	3.8 $\pm$ 0.8	$p = 0.0272$
UES-SF: Reward	4.5 $\pm$ 0.8	5.6 $\pm$ 0.9	$p = 0.011$

The qualitative data revealed that participants in the AR condition reported a greater sense of connection and interaction. Even without physical proximity, seeing where others were looking and engaging with shared virtual elements fostered social interaction. To enhance connection in remote multiplayer LBARGs, developers can consider including AR representations of remote players that show their position and also orientation.

4 CONCLUSION AND FUTURE WORK

This study presented a remote multiplayer AR hide-and-seek game evaluating the impact of AR on spatial decision-making, engagement, and player experience. The findings reveal that AR transforms traditional gameplay mechanics by introducing novel spatial possibilities and fostering collaborative interactions. Despite challenges such as navigation difficulties, this game design demonstrated higher engagement levels and unique spatial strategies compared to traditional co-located play, enhancing player experience and promoting deeper social connections across distances.

Future work needs to focus on incorporating dynamic features, such as enabling the hider to change hiding spots during gameplay, to further enhance spatial decision-making and collaboration. Such findings may inform the design of AR applications across other game genres and real-world scenarios, expanding the potential for engaging remote multiplayer LBARG experiences.

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